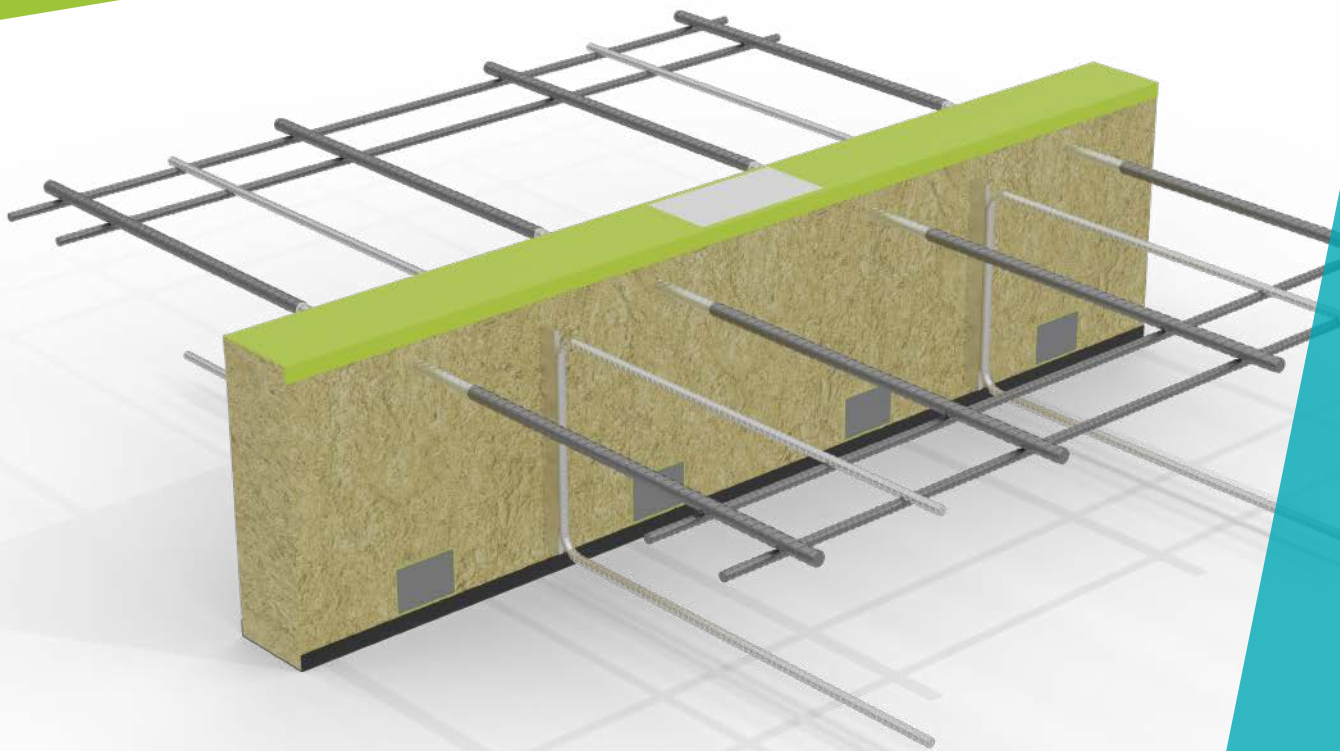


TECHNICAL MANUAL



EBEA® Balcony Connector

Load-bearing thermal break

Version PEIKKO GROUP 05/2020

EBEA® Balcony Connector

Load-bearing thermal break

- Effective elimination of thermal bridges for concrete balcony structures
- Efficient installation
- Robust construction with fire protection up to REI120
- Wide range of models for different applications

EBEA® Balcony Connector is a load-bearing insulated connection element for concrete structures which minimizes thermal bridges in cantilever balconies, or other applications such as walls and slabs.

Preventing thermal bridges and keeping structural integrity of cantilever balconies, walls and slabs at the same time might be a challenge. EBEA® Balcony Connector minimizes heat losses and prevents visual or structural defects to concrete structures. EBEA® system is very versatile in the wide range of models, you can find an optimal solution for transferring all transverse forces, horizontal forces and moments in relation to the main slab.

Noise reduction up to 21 dB can be achieved thanks to used special materials. A fire protection classification of up to REI120 is available with the EBEA® system.

Simple installation together with its features makes the EBEA® Balcony Connector a cost-efficient and practical solution for load-bearing structures with high requirements for minimizing thermal bridges.



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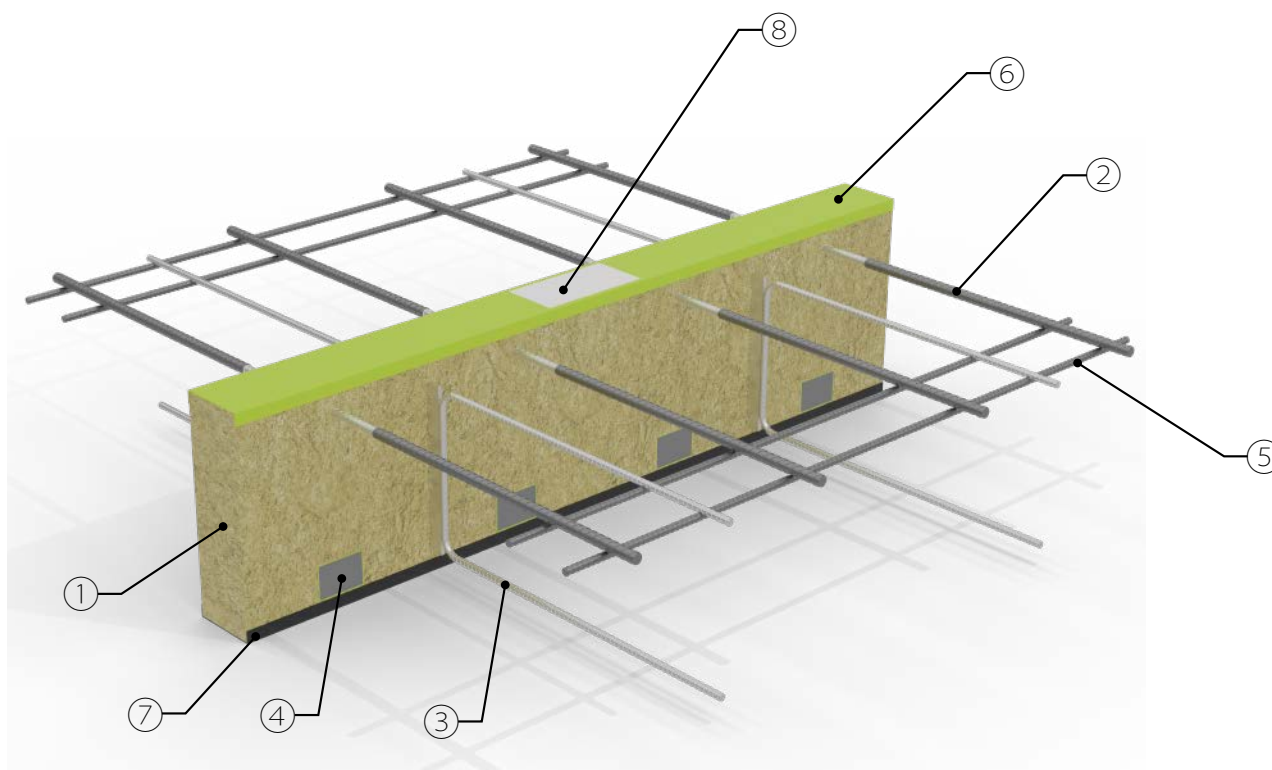
About EBEA® Balcony Connector

1. Product properties

The EBEA® Balcony Connectors are primarily used to eliminate thermal bridges in connection between cantilever balcony slabs and internal floor slabs. Balcony connector consists of thermal insulation, bending reinforcement, shear plate with anchor bars and accessories (see *Figure 1*).

The EBEA® Balcony Connector is cast together with the floor and balcony slabs or can be prefabricated with the precast balcony slab and attached to the cast-in-situ slab.

Figure 1. Description of EBEA® Balcony Connector parts.



- 1: Thermal insulation; 2: Bending reinforcement; 3: Shear plate with anchor bars; 4: UHPC compression buffer;
5: Cross bar; 6: Top cover - GREEN; 7: Bottom cover - BLACK; 8: Label

The EBEA® Balcony Connectors are available in various standard and semi-standard models that are suitable for a large variety of applications.

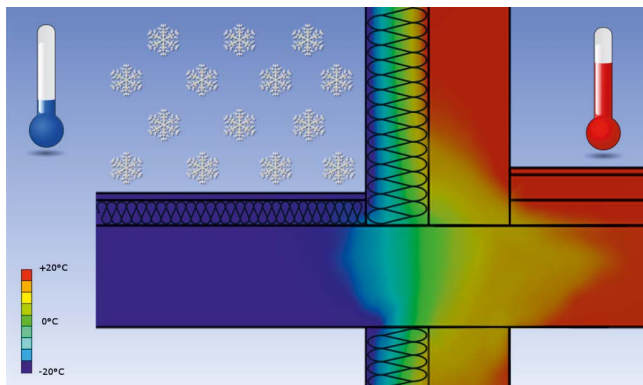
1.1 Thermal insulation

The high demand for improving the energy efficiency of buildings requires making the right design decisions already in the planning phase which will guarantee the reduction of thermal bridges and at the same time will allow for practical implementation.

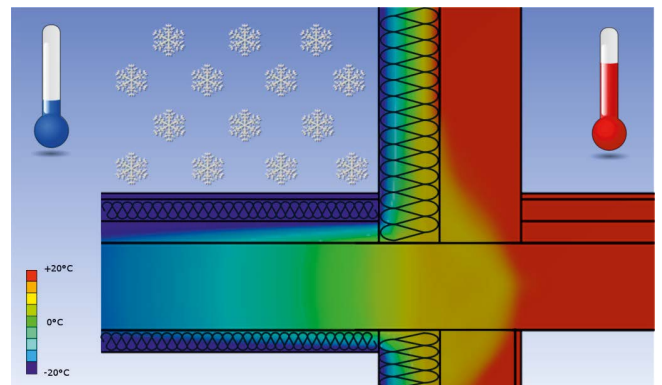
The aim is to prevent condensation and the subsequent mould formation caused by the cooling of the structure. By using the EBEA® Balcony Connector, these goals can be met, while preserving the load transfer and the stability of the structure.

The following illustrations show how EBEA® Balcony Connectors can improve reduction of heat transfer compared to other conventional solutions.

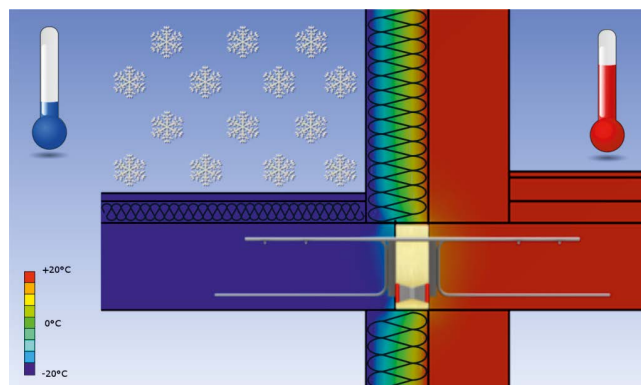
Solution without thermal insulation



Fully insulated balcony slab



Solution with EBEA® Balcony connector



1.2 Structural behaviour

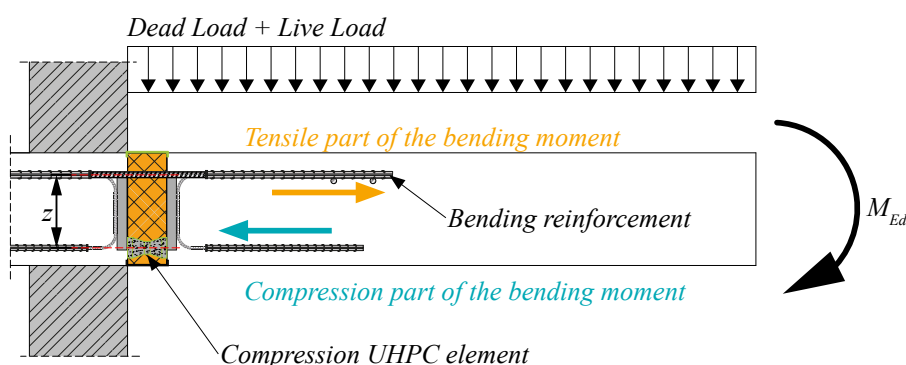
The EBEA® Balcony Connectors are designed to create a thermal break between two concrete elements and to transfer the vertical shear forces, bending moments or tensile loads through the thermal break.

Tensile forces caused by bending are transferred by the horizontal reinforcement placed on the top part of the balcony connector. Compression part of the bending moment is transferred by the buffers in the bottom of the balcony connector. Compression buffer is made either from UHPC or from stainless steel (Figure 2; a)).

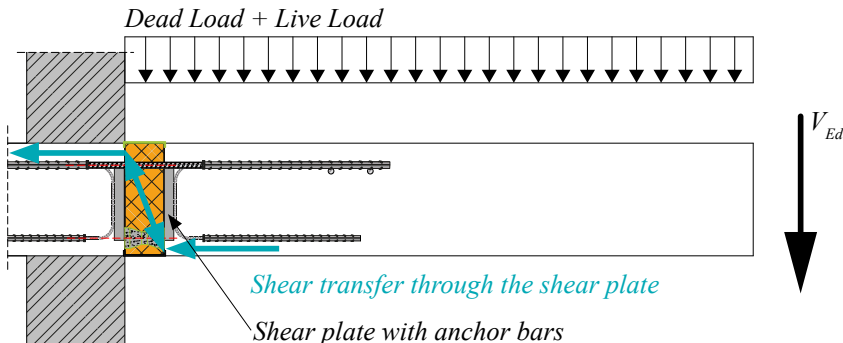
Shear loads are transferred either by vertical shear plates passing through thermal insulation or by inclined reinforcement in diagonal form (Figure 2; b)).

Figure 2. Load transfer in EBEA® Balcony Connector.

a) Transfer of the bending moment



b) Transfer of the shear load



The resistance values of the EBEA® Balcony Connector are determined by a design concept that refers to the Eurocodes (EN 1992-1-1, 2004), (EN 1993-1-1, 2005) and (EN 1993-1-4, 2006).

The load-bearing values specified in the resistance tables have been determined by assuming C25/30 concrete strength. In the case of the C20/25 concrete strength, load bearing capacity values need to be amended with 0.8 reduction factor.

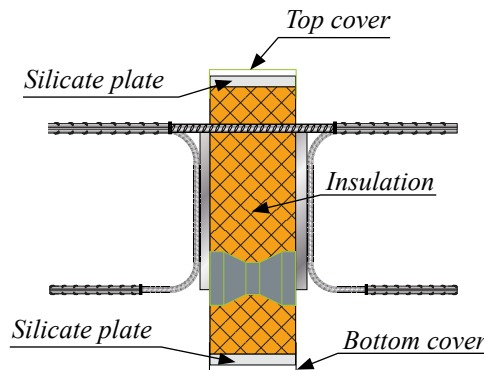
2. Fire resistance

The EBEA® Balcony Connectors with stone wool and integrated silicate plates (see *Figure 7*) creates fire resistant structure.

The fire test of EBEA® Balcony Connector was aimed to determine the fire resistance classes. In addition to the load-bearing capacity, the integrity and insulation of the elements was also tested.

The EBEA® Balcony Connector has fire resistance **REI120** if stone wool (SW) with silicate plates is used. EBEA® Balcony Connector has fire resistance **REI90** in case of use extruded polystyrene (XPS) with silicate plates and UHPC buffers.

Figure 7. Structure of insulation elements for fire protection requirements.



Short designations of fire resistance

- R** Load bearing; no loss of stability.
- E** Integrity; prevention of gas penetration on the non-flamed side.
- I** Thermal insulation; limitation of the transmission of fire or heat on the side facing away from the fire.



If EBEA® Balcony Connectors are installed discontinuously on REI balcony constructions, it is necessary to insert EBEA® spacers with corresponding REI classes in the intermediate areas.

3. Acoustic insulation

The **EBEA®-600 / EBEA®-1100 / EBEA®-1200 / EBEA®-1300 / EBEA®-1400** elements are acoustically optimized versions compared to models with shear plates. They are used when higher sound insulation is required. The following values are based on tests taken at the Lucerne University of Applied Sciences and Arts.

The tested elements have an 80 mm stone wool insulation and achieved the following sound insulation values:

EBEA® standard type	Sound insulation value ΔL_w [dB]
EBEA®-1100 4×10-1 D180 L1000 SW80	13.0
EBEA®-1100 6×14-4 D180 L1000 SW80	9.7
EBEA®-600-2 D180 L1000 SW80	21.5
EBEA®-600-5 D180 L1000 SW80	15.3
EBEA®-100 6×14-6 D180 L1000 SW80	6.2

The measured values given are purely informative and are to be understood only as a first indication and must be evaluated during the design of the envelope of the building. The sound insulation chosen can only be shown to be effective if the overall construction is tested by the building physics engineer or if the sound level is measured locally.

If you have any questions about the sound insulation value of other element types or insulation materials, please contact Peikko Technical support.